Tuesday, April 13, 2021 12:48 PM

$$CPI = \frac{\sum_{i=1}^{n} IC_i \times CPI_i}{IC} = \sum_{i=1}^{n} \frac{IC_i}{IC} \times CPI_i$$

$$\boxed{IC = \sum_{i=1}^{n} IC_i}$$

Fraction of instruction i

<u>Example 1</u> Suppose we have made the following measurements

- Frequency of FP operations = 25%
- Average CPI of FP operations = 4.0
- Average CPI of other instructions = 1.33
- Frequency of FPSQRT = 2%
- CPI of FPSQRT = 20 c.c.

• Assume that the 2 design alternatives are to

A. decrease the CPI of FPSQRT to 2

B. decrease the average CPI of all FP ops to 2.5

Which alternative is the best?

$$CPU_{original} = IC_{original} \times \underline{CPI_{original}} \times \text{Clock cycle time}_{original}$$

$$CPI_{original} = (0.25 \times 4.0) + (0.75 \times 1.33) \cong 2 \text{ c.c.}$$

$$CPI_A = CPI_{original} - (20-2) \times 0.02 = 2 - 18 \times 0.02$$

$$= 1.64 \text{ c.c.}$$

$= 1.64$ c.c.

$CPI_B = (0.75 \times 1.33) + (0.25 \times 2.5) = 1.625$ c.c

$$\text{Performance improvement}_B = \frac{\text{CPU time original}}{\text{CPU time}_B} = \frac{\cancel{IC} \times CPI_{original} \times \cancel{\text{Clock c.t.}}}{\cancel{IC} \times CPI_B \times \cancel{\text{Clock c.t.}}} =$$

$$= \frac{2}{1.625} = 1.23$$

<u>Example 2</u>

- We analyze 2 implementation alternatives for the conditional branches

- We have machine A with CPU_A, which has a conditional code set by a compare instruction, after which the subsequent instruction is testing the condition code => the branch is taken or not

- We have machine B with CPU_B for which compares are included in the branch instruction. Typically, we admit that at machine A, 20% of all instructions are branches. =>

if a=b then — 2
else — 1

CMP R1, R2
BNE addr

BNE R1, R2, addr

= 2

addr = 1

=> other 20% from the total no. of instructions are compare instructions at machine A

- We admit that the branch instructions require 2 c.c. and that all the other instructions require 1 c.c.
- We neglect the losses generated by memory system imperfection

- We neglect the losses generated by memory system imperfection
- Because compares are introduced in branches at machine B, the clock cycle duration at machine B has to be extended
=> admit that the clock cycle for B is 25% bigger than A.
- We are admitting (hypothetically) that all other organization aspects of the 2 computer systems are neglected

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$$CPU\,time_A = IC_A \times CPI_A \times \text{Clock cycle time}_A$$

$$CPI_A = 0.2 \times 2 + 0.8 \times 1 = 1.2 \text{ c.c.}$$

$$CPU\,time_A = IC_A \times 1.2 \times \text{Clock cycle time}_A$$

$$CPU\,time_B = IC_B \times CPI_B \times \text{Clock cycle time}_B$$

$$IC_B = 0.8 \times IC_A$$

$$\text{Clock cycle time}_B = \underbrace{1.25 \times \text{Clock cycle time}_A}$$

$CPI_B = ?$

$$\left.\begin{array}{ll} 20\% \ldots & 80\% \\ x\% \ldots & 100\% \end{array}\right\} \quad x = \frac{2000}{80} = 25\%$$

$$CPI_B = (0.25 \times 2) + (0.75 \times 1) = 1.25$$

$$CPU\,time_B = 0.8 \times IC_A \times 1.25 \times 1.25 \times \text{Clock cycle time}_A$$

$$= IC_A \times 1.25 \times \text{Clock cycle time}_A$$

=> A is better

Example 3

Example 3

We consider a load/store machine

Operation	Frequency	Clock cycle count
ALU ops	43%	1
Loads	21%	2
Stores	12%	2
Branches	24%	2/3

new

- 25% of arithmetic instruction use 1 operand from the memory

LDR R1, [R2]

ADD R2, R2, R1

- New machine for which, in the case of 25% arithm and logic ops are executed as <u>register-memory instruction</u>

=> allow 1 operand to be in the memory

new ALU op → 2 clock cycles CPI

new branch instructions require 3 instead of 2 c.c.

$$CPI_{old} = (0.43 \times 1) + (0.57 \times 2) = 1.57 \text{ c.c.}$$

$$CPU_{time\,old} = IC_{old} \times 1.57 \times \text{Clock cycle time old}$$

$$CPU_{time\,new} = IC_{new} \times CPI_{new} \times \text{Clock cycle time old}$$

$$\text{Clock cycle time}_{old} = \text{Clock cycle time}_{new}$$

$$CPI_{new} = (0.43 - 0.25 \times 0.43) \times 1 + (0.21 - 0.25 \times 0.43) \times 2 + (0.25 \times 0.43) \times 2 + 0.12 \times 2 + 0.24 \times 3$$

$+ 0.24 \times 3$

$1 - 0.25 \times 0.43$ $= 1.908$

$$IC_{new} = (1 - 0.25 \times 0.43) \times IC_{old}$$

$$CPU_{new} = IC_{old} \times \underbrace{(1 - .25 \times 0.43)}_{0.8925} \times 1.908 \times \text{Clock cycle time}_{old}$$

$$= IC_{old} \times 2.137 \times \text{Clock cycle time}_{old}$$

L/S mins! (Old!)